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Beyond the rationale of reduction: Exploring a deep ecological architectural practice of care

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ABSTRACT: This article explores how contemporary sustainable architecture may transcend the dominant *rationale of reduction* by embracing a deeper ecological sensibility through the ethic of care. The study examines the architectural project *Naturligrækkerne* by Danish office *Reværk*, developed under the *4>1 Planet* initiative for the *2023 UIA World Congress*. Drawing on Joan Tronto's moral framework of care, the aim is to analyze how care ethics can inform technology, material choices, and design processes. As such, the study advocates for a shift from an anthropocentric to an ecocentric worldview through concrete and critical practice, where relationality, adaptivity, and care responsivity offer new paths for sustainable architecture – an approach emphasizing the interconnectedness of humans and their environments in developing practices that are not only quantifiably sustainable but ecologically sensitive.

1 INTRODUCTION

This article intends to shed light on a blind spot in contemporary sustainable architectural practice, that is, the rationalist tendency towards a 'total quantification' and the deduced conception that strategies solely targeting the numerical reduction of abstract metrics like carbon footprints and LCA-values will be sufficient in restoring a stable ecological balance between the built and natural environment. Rather, the article will argue that the challenge of realizing 'truly sustainable' architecture requires a fundamental ontological turn away from the anthropocentric mindset from which our Western contemporary architectural practices ultimately descend.

This main argument rests on two theoretical positions. The first position understands the ecological crisis as ontological, rooted in a human-exceptionalist conception of the natural environment as a 'standing reserve' to growth and advancement. Identifying this ontological core uncovers the potential for an ontological turn; a turn away from the hegemonic paradigm of environmental abstraction, entailing a fundamental reconsideration of how to understand something as 'truly sustainable'. The second position concerns the possibilities of realizing such an ontological turn and maintains that our means of 'knowing' the world – the way we understand our interconnection with the more-than-human world – is mutually dependent on our means of interacting with it; what may be described as a reciprocal relation of influence between ontology and practice; actions and worldviews. An architecture will thus bear the mark of its time by manifesting and materializing the ontological position of its creation process. Yet, in a reverse fashion, grassroots activities, activistic interventions, and creative practices may equally inspire critical thinking that incrementally and accumulatively can lead to greater and more fundamental change. Within this second understanding thus lies the possibility of initiating ontological change through concrete practice.

Finally, introducing these positions into a Western contemporary architectural context, the article will examine how rethinking our ways of engaging with the environment through an enhanced qualitative and relational perspective – encapsulated by the *ethic of care* – may be a vehicle for enacting this proclaimed ontological (re)turn towards a worldview of deeper ecological sensibility. By seeking to move beyond the rationale of quantified reduction, it is *not* to dismiss the urgency for radically *reducing* anthropogenic pressure on the Earth's systems, but to rediscover our environmental interconnectedness that somehow has been lost during the efforts to modernize and rationalize built environments of the developed world.

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2 METHODOLOGY AND RESEARCH QUESTION

This article is divided into two parts. The first part will outline the theoretical framework to identify the ontological root of the ecological crisis and uncover the theoretical potential of inducing fundamental change directly through concrete and critical practice. Building on Joan Tronto's four elements of care, the second part of the article examines through an empirical case study the research question: *how may the ethic of care, as identified by Tronto's moral dimensions, inform material choices, technologies, and design processes in contemporary sustainable housing development to move beyond the abstract rationale of reduction?* The case of the study is an architectural experiment created under the initiative *4>1 Planet* leading up to the *UIA World Congress of Architects* in Copenhagen 2023, and the empirical data analyzed includes the project material created by the architects to accompany the final architectural experiment, photographic registrations, and project observations.

3 THEORETICAL FRAMEWORK

3.1 *Identifying the ontological root of the ecological crisis*

> *Modernity is, among other things, the triumph of technical prowess over nature. This triumph requires that nature be cleansed of transformative social relations; otherwise it cannot be the raw material of techne. (Tsing 2012, 513).*

In her article *On Nonscalability*, American anthropologist Anne Tsing uncovers how modern civilization, from a pre-industrial position to a global capitalist and extractivist state, has thoroughly depended on the extensive abstraction of Earth's natural environments. She describes this process as eliminating relational entanglements, diversities, and unique qualities to produce a 'scalable' version of nature, compiled of homogeneous quantities compatible with the rationality that characterizes modernity's mindset and practices. Descriptive for this mindset is the anthropocentric ontology; the ethical position that humans alone possess intrinsic value leaving anything non-human essentially to be considered a means to whatever ends humans deem necessary (Brennan & Lo 2022). When retracing the emergence of the ecological crisis, these abstracting tendencies stand central as common denominators shaping the Anthropocene means of 'knowing' the world, and it is here that we find the ontological root of the ecological crisis; a 'strange' human-non-human relationship characterized by abstraction, dualism, and alienation, ubiquitously represented in both extractive processes, where it continuously deepens the ecological divide, and in environmentalist endeavors as a tool for reconciliation. A somewhat ironic situation indicating the anthropocentric rationale's infiltration into both thinking and practice of Western society:

> *[. . .] contemporary environmental research and the environmental movement [. . .] appear to be increasingly dominated by a type of 'environmental accounting'. This abstraction and the rise of 'numerical environmentalism' have resulted in important, broader questions being foreclosed. (Smith 2018, 19)*

In *Sustainability, Wellbeing and the Posthuman Turn*, Thomas Smith thoroughly examines this 'ironic' situation, uncovering the often-imperceptible ecological implications behind otherwise well-meaning conceptualizations, such as the *ecological footprint* discourse and the *planetary boundaries* framework. His main argument maintains that abstract numerical environmentalism obstructs deeper engagement with the dysfunctional relationship between humans and the non-human world. For example, we see the planetary boundaries framework promoting global-scale solutions like geoengineering strategies, while the ecological footprint discourse encourages individualistic actions focused on reducing numerical metrics, resulting in the outsourcing of practices with 'dirty' footprints (Smith 2018, 26). In both cases, the strategical outcomes are abstract

coping mechanisms that fail to recognize diversity and qualitative nuances, reinforcing, instead, an anthropocentric dualistic notion of an abstract natural environment 'out there'; something we may harm through action yet remain decoupled from our everyday lives (Smith 2018, 27).

Neither Smith nor this article intends to dismiss scientific efforts to quantify environmental markers and thresholds. Knowing the 'state' of the Anthropocene, even in abstract manners, is a valuable insight that may not simply reveal our most critical ecological impacts but also inform, unite, and incentivize global, local, and individual action. Yet, the point raised above is that strictly relying on the quantitative paradigm in developing our ecological practices – to 'aid the planet' by distancing ourselves from it – is a strategy bearing the risk of strengthening the very same mindset that produced the ecological imbalance in the first place. If we consider, from this position, the rationale of reduction, it may be seen as having a double negative impact; firstly, by being a symptomatic treatment solely addressing the consequences of the underlying environmental abstraction and therefore incapable of facing the problem at the ontological level; and secondly, by potentially standing in the way of more fundamental change by advancing through abstractions the same anthropocentric mindset from which it originates, consequently obstructing strategies operating beyond the numerical paradigm. This is the 'blind spot' of contemporary sustainable architecture: the inability to conceive the intricate interrelation between the ecological crisis, our anthropocentric position, and the environmentalist practices that this position produces. However, as Tsing reminds us, this influence is reciprocal:

> *Conceptualizing the world and making the world are wrapped up with each other – at least for those with the privilege to turn their dreams into action. The relationship goes both ways: new projects inspire new ways to think, which also inspire new projects." (Tsing 2012, 506)*

As such, within this codependency is also the possibility of more fundamental change; a reversed path arising from the bottom-up, providing each practitioner the capacity to enact this ontological turn through local, concrete, and critical action directly. This path will be explored in the remainder of this article.

3.2 *Care as an ecological strategy*

Today, the anthropocentric mindset is ubiquitous, pushing its mathematical rationale into every corner of modern Western society. The truth is, however, that the complexity of the real world far exceeds what any abstract model can ever render. Yet, the insistence on *unambiguous certainties* within Western contemporary building practices as seen in recent sustainability trends such as *the net-zero discours*e (World Green Building Council 2019), the notion of *absolute sustainability* (Ryberg et al. 2020), or the Danish *Reduction Roadmap* initiative (Reduction Roadmap 2024) has motivated the negligence of this simple truth. Thus, to turn away from the anthropocentric position is to challenge present categorical sustainability conceptions, welcoming a more uncertain, complex, and qualitative worldview – one where the built and natural environment is seen as intricately interconnected and not as isolated and solid objects frozen in time. Naturally, this is a confrontation with the modern tradition in which architectural ideals are clean, static, and universal – yet, more profoundly, it is a confrontation with the mathematical rationale that has guided this modern paradigm and its environmental perspective – both of which still prevail today.

In the search for an alternative path or guidance compatible with this post-anthropocentric position, the ecocentric worldview offers a relevant foundation. Contrary to anthropocentrism, where environmental action essentially seeks to conserve instrumental value for humans foremost, ecocentrism places intrinsic value in everything living, human and non-human alike, maintaining that true well-being depends on the well-being of the vast networks of ecological and social relations that make up our worlds (Brennan & Lo 2022) – a distinction solidified in 1973 by Norwegian philosopher and ecologist Arne Næss' conception of the 'shallow' and 'deep' ecological approaches. Whereas the 'shallow' ecological approach takes a utilitarian orientation, such as fighting pollution and resource depletion for the benefit of the population in developed countries, 'deep' ecologists are concerned with fundamental principles of diversity, complexity, autonomy,

decentralization, symbiosis, egalitarianism, and classlessness (Naess 1973, 1). What characterizes the deep ecological or ecocentric position is, therefore, an opposition to the abstracting and dichotomizing tendencies on which modernity has been erected – an opposition situated in a strong ethical foundation emphasizing context, relations, and qualities over abstract concepts and universalized principles; an ethical foundation many scholars have later tied to the moral theory of care ethics (Smith 2018; Dalziel 2022; Rawes 2013; Tronto 2019).

Care ethics has its roots in the mid-80s field of psychology and feminism. Normatively, care is considered a *practice* encompassing the actions we take to maintain, continue, and repair our 'world' understood as the life-sustaining web that includes ourselves, our relations, and our environment (Tronto & Fisher 1990). As a moral theory, it is attributed to American psychologist Carol Gilligan's work exposing gender bias within moral approaches and justice perspectives, uncovering through her studies how women would attend to specific circumstances, relations, and responsibilities in moral dilemmas, in contrast to men who, more often, would rely on universalized principles and rules (Gilligan 1982; Sander-Staudt 2011). Since then, many scholars have elaborated and extended the moral theory of care, also broadening it from its strong ties to gender studies to other fields and disciplines including environmental ethics (West et al. 2020). Notable, in the context of this article, is Joan Tronto who identifies four elements of care and subsequently how these four elements constitute four corresponding moral dimensions that may be of relevance when reconsidering architectural practice from a post-anthropocentric position:

Caring about, the element of care that involves recognizing that care is necessary and assessing whether and how this need should be met (Tronto 2020, 106). Its moral dimension, *attentiveness*, describes the need to be attentive to the needs of others in a caring relation (Tronto 2020, 127).

Taking care of, the element of care that describes how assuming responsibility for the caring process necessitates determining how this need should be met and recognizing that it can be done. If one *caring about* does not believe anything can be done to meet the need of care, then there can be no *taking care of* (Tronto 2020, 106). Its moral dimension of *responsibility* should not be mistaken for the notion of *obligation,* as *obligation* relates to conventional societal and cultural norms, formal agreements, duties, and legal bonds, while *responsibility* relates to the concrete qualities of a situation and takes on a more ambiguous and flexible connotation that necessitates its frequent reevaluation (Tronto 2020, 132).

Caregiving is the element of care that involves the physical process of meeting the needs of care and requires, almost always, direct contact between the *caregiver* and the objects of care (Tronto 2020, 107). *Competence* is the moral dimension of *caregiving* and concerns that the actual need for care is met adequately (Tronto 2020, 133).

Care-receiving is the final element of care and encompasses the care-receiver's response to the care given. Thus, to be cared for adequately relies on a responsive or communicative aspect to provide certainty that the actual need for care has been met. Its moral dimension, *responsiveness,* does not equal the ethical concept of *reciprocity*. Whereas *reciprocity* is about putting oneself in the situation of others, *responsiveness* requires one to consider the situation of others as the other expresses it (Tronto 2020, 136).

Reconsidering sustainable practices from the ethic of care outlines a 'deeper' ecological perspective operating beyond the *rationale of reduction*. At this point, distinguishing the natural from the built environment becomes counterintuitive as these will be conceived as inseparably interconnected. Instead, a turn like this will relativize how we conceptualize and understand the notion of something 'truly sustainable': where the anthropocentric paradigm may look to conceptual frameworks for a universal, categorical, and quantifiable definition to project onto the environment (built or natural), the ecocentric approach, guided by care, will take an opposite direction, being foremost attentive to the specific qualities and relations that a given situation provides, taking care to take care for everything and everyone affected.

4 CASE STUDY: CARE & ARCHITECTURE

"Rather than thinking of buildings as things, thinking of them in relationships with ongoing environments, people, flora and fauna—that exist through time as well as in space, changes the approach fundamentally." (Tronto 2019, 28)

4.1 *The case*

The project under investigation was developed in the run-up to the *UIA World Congress* in Copenhagen 2023 under the initiative *4>1 Planet* and the *Next Generation Architecture* competition which intended to demonstrate new perspectives of sustainable housing practices. In the aftermath, the project received broad media coverage specifically celebrating its different approach to materiality and building technology against the mainstream of contemporary architecture. It is a temporary project, demonstrating through an experimental pavilion construction principles for a yet un-realized row-house project *Naturligrækkerne* by the Danish architectural office *Reværk* (Reværk 2023). Being a temporary structure, its contextual relationality has not been emphasized by the architects, and will, therefore, not be investigated in the following. Further, conceived as an architectural experiment, the project development was not in the same way restricted by the hindrances and obstacles that otherwise preoccupy the 'real world' – an opportunity that provided the architects freedom to explore the more radical possibilities of materiality and technology. As such, the development of the project followed three lines of material exploration with an emphasis on identifying and utilizing in the best way possible each material's unique properties regarding, among others, maintenance, weathering, protection, possibilities for disassembly, fabrication, recyclability, indoor climate, and carbon footprint. In what follows, these three lines of exploration will constitute the three parts of the case analysis.



Figure 1. Details of the pavilion. Photo: Reværk Arkitektur.

The first line of material exploration is *raw rammed earth* used to construct the interior walls dividing the housing units (Reværk 2023). The qualities highlighted by the architects were respectively; its capacity to absorb and release moisture from the air, thus regulating indoor air quality; its capacity as thermal mass, securing a stable indoor temperature; its low carbon footprint; its local availability; its fire-retardant capacities; and its acoustic and sound absorption qualities, making the material ideal for dividing the different housing units. Further, a particular focus was given to the optimization of the construction process considering the use of robotic arms in the ramming of the earth, and it is stated in the project description that automatizing this process is both realistic and may drastically reduce costs making this rather labor-intensive material available in more conventional practices (Reværk 2023, 4). Given that it is a *raw* earth wall, in the sense that no cement binding has been applied to the material, its construction is equally as simple as its deconstruction; applying pressure solidifies the material and it can as easily be dissolved back into its raw form of clay and soil. To protect the construction, the architects emphasized its interior position while, in the constructed pavilion, protecting the ends of the wall exposed to the exterior with a large roof overhang. The expressive S-shape provides stability to the rammed earth construction. At the same time, the middle section positioned parallel to the roof windows is exposed to sunlight throughout the day, thus utilizing its capacity as a thermal core.

The second line of material exploration is centered around the use of *wood* (Reværk 2023). In the project, the architects utilized traditional principles for timber construction rationalized through modern processing and construction techniques. Keeping the timber construction internal and exposed enabled continuous observation of the material condition which allowed

for easy maintenance, repair, and potential disassembly in case necessary. Using primarily round timber elements, a higher utilization rate of the unprocessed harvested timber was possible, minimizing waste products and cut-offs. According to the architects, this higher degree of utilization may potentially aid in reducing the high ecological impact of conventional timber production, for example, clear-cutting practices where entire fields of uniform trees are cut down, leaving behind a devastated ecosystem (Reværk 2023, 5). In addition, the round timber element adds a distinctive expression to the project, underlining the reference to traditional timber details – especially in the load-bearing construction where the necessary dimensioning of the round timber also appears lighter compared to conventional saw-cut timber.

The third line of material exploration concerns the insulation and cladding of the project (Reværk 2023). What informed this design process was the need to protect the underlying timber construction without preventing its maintenance, repair, and potential disassembly. The solution was to develop prefabricated insulation cassettes designed to be independently installed from the load-bearing construction. This also allowed for the easy maintenance of both cassettes and timber construction. Wood fiber insulation secured the construction's breathability, which, combined with the internal rammed earth wall, improved the indoor climate. This also supported the conditions for the timber. In addition, wood-fiber insulation has a significantly lower carbon footprint compared to conventional mineral-based materials. In this regard, the architects emphasized that by using a thoroughly tested product of a well-established manufacturer, the realizability of the project is aided. The cassettes themselves were designed to require the least amount of solid wood necessary with as few joints as possible. Internally, a wood board cladding with a paper-based 'vapor brake' underneath is installed on the cassettes, and externally, lime plaster is added to the façade. Finally, an emphasis was put on designing the details of the cassette installment, allowing for potential disassembly. The resulting simple joint utilizing wood dowels enhances the aesthetic expression of the raw timber construction.

4.2 *An architecture of care?*

All in all, these three lines of material exploration encapsulate the primary 'rationale' of the project. Although the architects themselves haven't articulated this 'rationale' as being guided by the ethic of care, many aspects within each design choice can be analyzed through this ethical lens. Starting with Tronto's first element of care, *caring about*, and its moral dimension, *attentiveness*, we may ask what specific needs the architects *cared about* and *attended* to. The simple answer here is, of course, that there is no simple answer and that any design process is an intricate and multi-dimensional maze of choices, priorities, and considerations. Yet, what *is* apparent is that the priority foremost lies in the agency of the materials, and only by being *attentive* towards this agency did the architects begin to give form to the architectural project overall. Thus, where a 'modern' approach conventionally would start by defining the necessary space program *attentive* to whatever ideal we may conceive and then subsequently subject the materials to this rationale, the approach in the project is characterized by investigating how *we* may best subject *our* inhabitation to the materials and *their* specific caring needs. In this manner, it is an opposite approach, *attentive* – not to the conventions of a modern housing standard or sustainability paradigm – but to what may be described, using Tronto's term, as the *responsiveness* of materials. Against a modernist rationale of human-first orientation, we may understand this *attentiveness* as ecocentric given its *caring about* the intrinsic qualities of the materials in the project and how they relate to one another in constituting an architectural environment that may best shelter – not only the human inhabitants – but the concrete physicality constituting the project itself.

Another 'rationale' we may analyze through Tronto's framework relates to the architect's choice of materials. In the initiative behind the project, '4>1 Planet', there is, as the name suggests, an embedded *rationale of reduction* referring to the Danish building industry's environmental footprint currently being four times larger than what the Earth can sustainably support. As this reduction rationale became a condition for the competition entries, it may be assumed that it somehow influenced the architect's deciding which materials to work with. Yet, this rationale aside, considering the architect's profile and portfolio of projects it is evident that the choice of working with rammed earth and biogenic materials is naturally in line with their general architectural methodology indicating that a different motivation may also be present. From a care ethical

perspective, we may try to understand this motivation using Tronto's second moral dimension as an ecological *responsibility* to *take care of*. As such, by considering the local availability of the rammed earth, the environmental implications of timber production, the potential of using round timber, or the biogenic qualities of wood fiber insulation, the architects demonstrate a *responsibility* for the wider ecological web of life they operate within. Turning our attention toward the concrete project, however, a different scale of *responsibility* becomes evident, centered around the caring processes of the architecture itself and how this best can be met. By choosing materials and technologies capable of being maintained and prioritizing these maintenance processes in the design choices, such as leaving the timber structure exposed or visually emphasizing the dowels securing the insulation cassettes, the architects celebrate and encourage the *responsibility* for *taking care of* the local ecosystem that is the specific architectural environment of the project. An interesting design choice in this relation is the raw rammed earth wall, which *will*, given its material properties, release dust and sediments over time to the interior. Whether this should be seen as an artistic quality or an annoying practicality, it can undoubtedly be seen as a material *response* that motivates a *responsibility* for *taking care of.*

Yet, from a care ethical position, another material emphasis can be identified using Tronto's third and fourth elements of care, namely *caregiving* – its moral dimension *competence* – and *care-receiving* – its moral dimension *responsiveness*. Again, *competence* is about ensuring that the specific caring need is met adequately, and it is by being *attentive* to the material *responses* that adequate *competence* can be identified (Tronto 2020, 133). This adds another dimension to the architectural practice of care that makes demands on the material properties and technologies utilized: for a material to engage in a caring relation it must be *responsive*. As such, embedded within any material is knowledge about its caring needs, however, only materials with *responsive capabilities* may allow its implicit knowledge to be recognized and *attended* to by a caregiver – and only then may *responsibility* be taken to *take care of* leading to *competent caregiving.* If a material fails to be *responsive*, a caring relation may not be possible at all, or at least be attempted blindly and without *competence* if not the caregiver already possesses knowledge about the specific material by being a specialist – and even so may proper *responsiveness* necessitate out-of-the-ordinary equipment. Take a concrete wall, for instance, its solid permanence –would a non-specialist be able to uncover the embedded knowledge behind this barrier of non-responsiveness or see it simply as reluctant to care-receiving? Reconsidering, against this example, the rammed earth wall, we may conceive its continuous state of change as *responsiveness*, motivating foremost, as mentioned above, *responsibility* for *caretaking*; yet, more than that, as an exchange of embedded knowledge informing adequate caregiving and, therefore, how a sustainable practice may operate in the given context. Or the lime plaster of the cassette exterior; it may be touch-sensitive, change color over time, and tint the jacket of one leaning against it, but it is within these responsive capabilities that knowledge can be uncovered to advise the caregiver about the practices necessary for adequate environmental care and maintenance. Thus, by emphasizing this material responsiveness in the design choices, the architects approach this architectural perception that Tronto herself promoted, thinking of buildings not as things, static and bound to three-dimensional space, but as interconnected parts of our ever-changing ecological environments, spanning both space and time. This is a 'weak' architecture – weak in the sense of being perceptible to the influences of a world in flux. Compared to contemporary modernism's ideal of static steel, glass, and concrete, this weakness may easily be mistaken for fragility. However, this weakness is not understood in any bad sense. It simply means that it is care-dependent, and its durability is achieved, not through material strength and permanence, but through the ongoing *practice of care.*

5 CONCLUDING REMARKS

This article has aimed to examine how an enhanced ecological sensibility in architectural practice, framed by the ethic of care, may contribute to a fundamental shift away from the anthropocentric worldview to which we may trace both the ecological crisis and the modern project of the Western world. By achieving a better understanding of how our worldviews and actions are intricately interconnected, the article intended to uncover the possibilities for enacting ontological change through direct, concrete, and critical practice. In the case study, emphasis was limited to the architect's

considerations concerning materiality and technology. As these are just two of many architectural agencies, the case study should not be mistaken for a comprehensive examination of how architecture in full complexity should be conceived from the ethical perspective of care. Rather, the intention has been to demonstrate how *care,* as a moral foundation, may guide architectural practice beyond the predominating *rationale of reduction*. Because the focus of the case study was not on the wider context of the project but centered around qualities that may be conceived as less contextually dependent, meaning the inner 'locality' of the project itself, its materials, and how these relate to each other, it can be argued that an important relational depth is missing in the above study. As this may be true, it should be noted that any practice informed by care *will* be contextual and qualitative, making the definition of more generic design principles challenging and arguably a counter-intuitive endeavor. Yet, by focusing on the architectural qualities that are not defined through the project's wide contextuality but the very internal ecology of the architecture, it is hoped that findings and reflections may still hold relevance in a different situation than the one studied. That being said, to challenge the dominating *rationale of reduction* through a qualitative sensibility guided by the ethic of care *is* a strategy that may be projected onto the broader field of architecture without paradoxically defining abstract and universalized ideals in stark contrast with the very core of care ethics. As such, it is not new solutions but new practices being discussed and proposed: practices whose architectural manifestations – following the ethic of care – will be responsive, malleable, adaptable, and, in that sense, *weak*. Giving up the illusion of permanence in favor of this 'longevity of weakness', where durability, relevance, and functionality are subjected to and reinvigorated through the ongoing practice of care, is perhaps the most pronounced architectural outcome of this above-proposed ontological *turn*. It frames a different sustainable perspective, to which the relation between the caregiver and the responsive environment becomes central. To fully uncover the sustainable perspectives of moving beyond the *rationale of reduction,* this relationship needs to be further explored in both private, public, and professional settings. For now, however, we may argue that from a perspective of care, a sustainable future is not something we invent and subject the world to. It is the other way around; it is knowledge already embedded in our environments about how best to care for and maintain a habitable world for both living and non-living alike. The knowledge is out there, somewhere behind the abstractions of the anthropocentric rationale – being attentive to it remains the challenge.

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